

Authentication of administrative documents using Blockchain technology

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EAI AfricaTek 2026 - Ouagadougou, June 2026

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Introduction

- Administrative documents are very important for government agencies and the public.
- The reliability of certain administrative documents is increasingly controversial, as it is now common to find falsified documents or documents of dubious origin in the public digital space [1] [2].
- Blockchain technology offers an immutable, decentralized, and transparent ledger [3] [4] [5] — a promising foundation to reduce the risk of falsified documents.

Main question

How can blockchain technology be used to simplify the verification of the authenticity of administrative documents and reduce the risk of falsified documents being used ?

Objectives

1st

To propose a hybrid decentralized approach to authenticating administrative documents based on the non-repudiation, high transparency and decentralized storage properties of Blockchain.

2nd

To propose a hybrid DApp using the Ethereum blockchain, digital signatures, hashing, electronic timestamping, and smart contracts to enhance document security.

State of the art (1/4)

- Administrative documents = “documents produced or received, in the context of public service, by the State, local authorities and other public or private entities entrusted with such a mission;” [6].

Omar S. Saleh et al. (2020) [7]

A framework based on Hyperledger Fabric for verifying educational certificates in order to address the issue of forged university degrees.

G. Malik et al. (2019) [8]

A unique document verification model to resolve the tedious and time-consuming process of document verification.

State of the art (2/4)

F. Azzam et al. (2023) [9]

A system model called SECHash integrating the LBRY's network and OCR technologies with the aim of maintaining the integrity and accuracy of incoming invoice data by prohibiting modification after acceptance by government agencies. It also ensures that accepted invoices are neither destroyed nor lost.

R. F. Ghani et al. (2022) [10]

The main advantages of applying blockchain technology to electronic administrative systems include: the ability to store and share government, citizen, and business data in a secure and controlled manner; reduced management costs and risks of corruption/abuse; the elimination of human intermediaries; and increased trust among users of e-government service.

State of the art (3/4)

I. Meirobie et al. (2022) [11]

A framework for authenticating electronic documents using Blockchain technology for the Indonesian government system. Go-Chain based on the hashing (SHA-256), signature (P-256 curve DSA) of the digital transcript of the document and the PoA consensus protocol.

P. Mishra et al. (2025) [12]

A framework that integrates a permissioned Ethereum-based blockchain with PoA as the consensus mechanism. The approach incorporates the use of SHA-256 hashing, timestamping, and smart contracts. On 1,000 PDF docs, the proposed framework resists man-in-the-middle attacks and detects 93.1% of shadow attacks with 98.4% accuracy and only 1.6% false positives, unlike traditional PKI methods, which detected no forgeries.

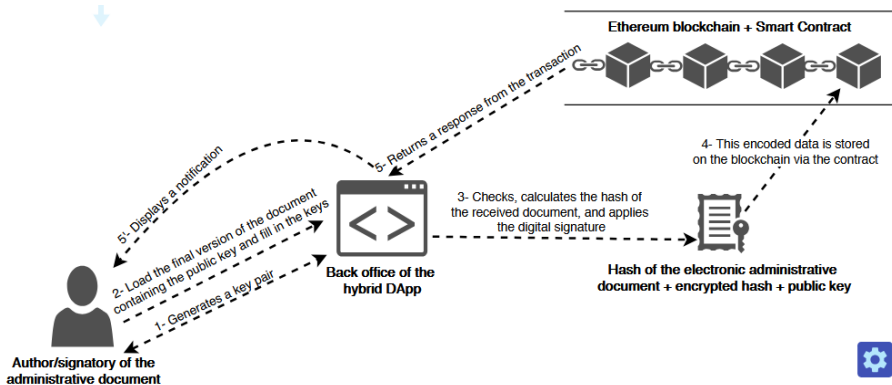
State of the art (4/4)

Table: Summary of existing methods.

Authors	Technologies	Limitations
Omar et al.[7]	Hyperledger Fabric.	Academic degrees only.
Azzam et al.[9]	LBRY's network, OCR, PostgreSQL.	Invoices only.
Ghani et al.[10]	Java TCP socket, Blockchain.	Scalability, Interoperability.
Mishra et al.[12]	SHA-256, timestamping, smart contracts, permissioned blockchain.	PDF support only.

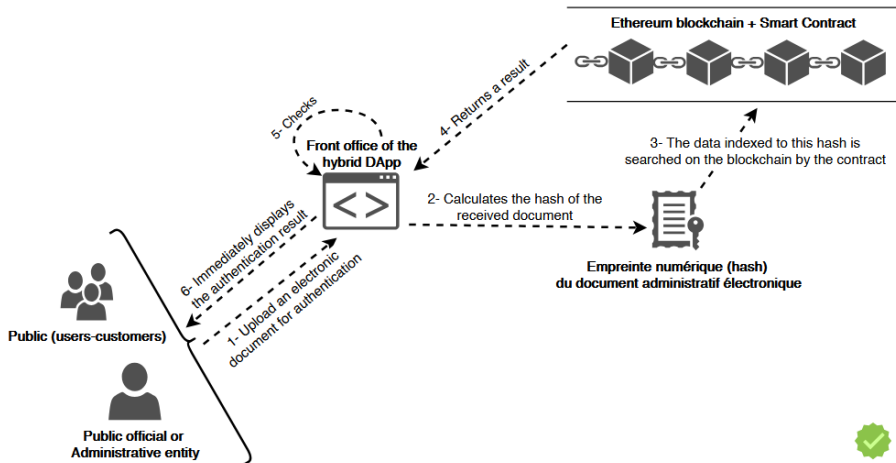
Contribution : Rapid verification of official documents using a hybrid approach that combines Ethereum and smart contracts, timestamping, digital signatures, and hashing.

Methodology (1/2)



Phase 1 - Recording administrative documents in the Ethereum blockchain.

Methodology (2/2)



Phase 2 - Verification or authentication of administrative documents on the Ethereum

Implementation & Discussion (1/5)

- For approximately 94.2 Kb, the following would be required on average :
 - 1.76 millisecond for an authentication attempt of a document (not previously registered on Ethereum);
 - 4247.66 milliseconds (4.24 seconds) and 0.0005168205083614 ETH to register the document on Ethereum;
 - 136.12 milliseconds (0.13 second) for authentication of the administrative document.
- The proposed system enables near-instantaneous document authentication.

Challenges :

- interoperability between the Ethereum blockchain network and existing government infrastructure;
- adoption of blockchain in e-government.

Implementation & Discussion (2/5)

- JSON Remote Procedure Call (JSON RPC) protocol
- RESTful API
- Ethers.js
- SHA-256 algorithm
- secp256r1 (P-256)

Implementation & Discussion (3/5)



ethereum



Hardhat



Implementation & Discussion (4/5)

Enregistrement de document administratif sur Ethereum

Charger le fichier .crt de clés

+ Chargez le fichier .crt de clés

trusted-bu
ndle-17474
96401116.c
rt

217 B

X

Clé privée du signataire

Renseignez votre clé privée cryptographique

Clé publique du signataire

Renseignez votre clé publique cryptographique

Document numérique (PDF ou Word)*

+ Chargez le document administratif (PDF/Word)

Communiq
ue_test.do
cx

12.284 KB

X

✓

Enregistrer le document sur Ethereum

X

Effacer le formulaire

Résultat de l'enregistrement

Statut de la transaction: Succès (Transaction minée)

Document soumis: Communiqu_e_test.docx

ID transaction: 0x21b1fd5b23b850e4bac6b297a03b2ed5d4f8821fc1b496cedead721385095aab

Type transaction: (2) EIP-1559 : transaction (standard recommandé)

N° Bloc: 2Nonce: 1

Signataire (EAO): 0xf39Fd6e51aad88F6F4ce6aB8827279cFfB92266

Adresse du contrat intelligent: 0x5FbDB2315678afecb367f032d93F642f64180aa3

Gaz total utilisé pour le bloc: 322200

Gaz réellement consommé pour la transaction: 322200

Prix réel du gaz payé pour la transaction (Wei): 570596726554200

Prix réel du gaz payé pour la transaction (ETH): 0.0005705967265542

Store administrative documents.

Implementation & Discussion (5/5)

☑ Authentification de documents administratifs à l'aide d'Ethereum 2.0

+ Chargez le document administratif (PDF/Word)

COMMUNIQUE_01.pdf109.883 KB

✕

Vérifier l'authenticité du document

Réponse d'authentification de document administratif -Ethereum 2.0

✓ Le document est authentique

Document vérifié :COMMUNIQUE_01.pdf

Le contenu du document est-il intègre ?

✓ Oui

Date de la demande :14/01/2026 12:47:46

Horodatage blockchain :1768394068 | mercredi 14 janvier 2026 à 12:34:28

Empreinte numérique du document :EFAIgoDeMLP14zgCY4eVREg9D4FpRO6bF8DjciRVE=

Clé publique associée :MFkwEwYHKeZizjDCAQYIKoZlZjODAQcDQgAEIglppDWtCzL+LlCGGwR7OnDoGUIX3QXCWXdKkbmzW3+z71CSDGRZQgOdmtYc+i0mwmdgWPrXjDEzZp03I2nQ==

Type de clé :ECDSA

Courbe elliptique :secp256r1 (P-256)

Verify or authenticate document.

Navigation icons

Conclusion & Perspectives

Contributions

This study confirms that blockchain technology (Ethereum) offers promising potential for ensuring the authenticity and security of administrative documents.

Future perspectives

The need to address the concern of :

- how the proposed system will function in a non-peer-to-peer network such as that of the current administration,
- how to adapt the public expenditure system to take into account gas fee payments.

Acknowledgement

Thank you.

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